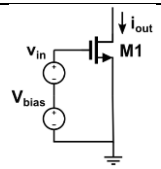
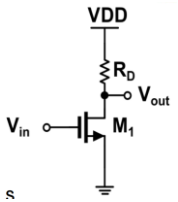
# Common Source




## Large-Signal Analysis

### With Resistive Load




Saturation: $g_m = \mu_n C_{ox} \frac{W}{L} R_D \left(V_{GS} - V_{TH}\right) = \mu_n C_{ox} \frac{W}{L} R_D \left(V_{in} - V_{TH}\right)$

Linear: $g_m = \mu_n C_{ox} \frac{W}{L} V_{DS} = \mu_n C_{ox} \frac{W}{L} V_{out}$

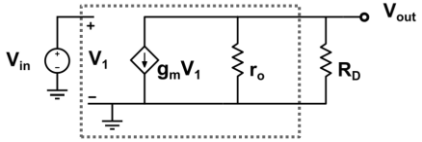
$$A_v = -g_m R_D = -\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \frac{V_{RD}}{I_D} = -\sqrt{2\mu_n C_{ox} \frac{W}{L}} \frac{V_{RD}}{\sqrt{I_D}}$$

Channel-Length Modulation

$$V_{out} = V_{DD} - \frac{1}{2}\mu_n C_{ox} \frac{W}{L}\left(V_{in} - V_{TH}\right)^2 \left(1 + \lambda V_{out}\right) R_D$$

$$r_O = \frac{1}{\lambda I_D} \quad A_v = \frac{\partial V_{out}}{\partial V_{in}} = -g_m \frac{r_O R_D}{r_O + R_D} = -g_m \left(r_O \parallel R_D\right)$$

## Small-Signal Analysis




$$V_{out} = -g_m V_1 \left(r_O \parallel R_D\right)$$

$$A_v = -g_m \left(r_O \parallel R_D\right) = -g_m \frac{r_O R_D}{r_O + R_D}$$

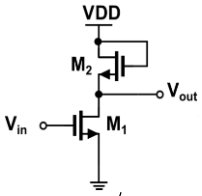
$$= -g_m R_{out}$$

$R_{out} = \left(r_O \parallel R_D\right)$ if $\lambda$=0, $r_o$=∞ $R_{out} = R_D$

$R_D \to \infty$ $A_v = -g_m \left(R_D \parallel r_O\right) = -g_m r_O$

### With Diode Connected

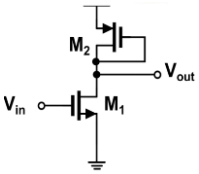
NMOS



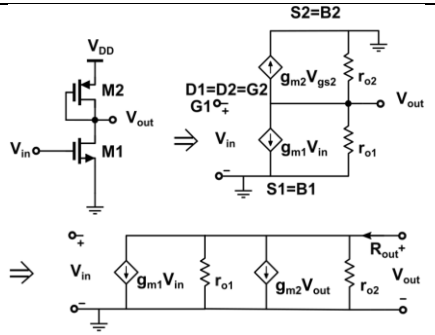

$$A_v = -\sqrt{\frac{2\mu_n C_{ox} (W/L)_1 I_{D1}}{2\mu_n C_{ox} (W/L)_2 I_{D2}}} \frac{1}{1+\eta} = -\sqrt{\frac{(W/L)_1}{(W/L)_2}} \frac{1}{1+\eta}$$

$\eta = g_{mb2}/g_{m2}$, With Body-Effect

PMOS




Without Body-Effect

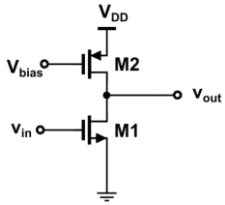


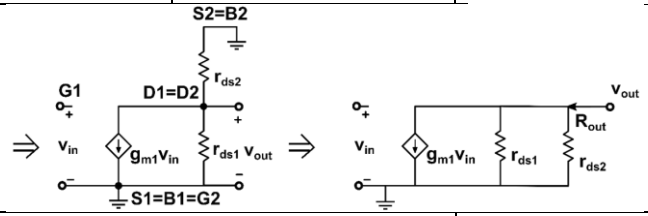

$$R_{out} = \frac{1}{g_{O1} + g_{O2} + g_{m2}} = \frac{1}{g_{m2}} // r_{O1} // r_{O2} \approx \frac{1}{g_{m2}}$$

$$A_v = \frac{-g_{m1}}{g_{O1} + g_{O2} + g_{m2}} \approx -\frac{g_{m1}}{g_{m2}} = -\sqrt{\frac{\mu_n (W/L)_1}{\mu_p (W/L)_2}}$$

$$A_v = -g_{m1} R_{out}$$

### With Current Source Load

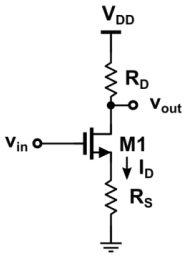



$$R_{out} = \frac{1}{g_{ds1} + g_{ds2}} \cong \frac{1}{I_D(\lambda_1 + \lambda_2)}$$

$$A_v = -g_{m1} R_{out} = -g_{m1}\left(r_{o1} \parallel r_{o2}\right)$$

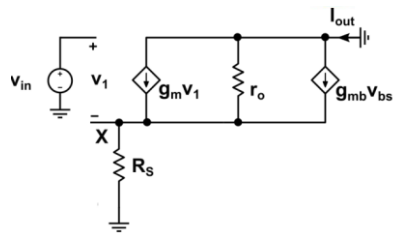
### with Source Degeneration




$$A_V = \frac{\partial V_{out}}{\partial V_{in}} = -\left(\frac{\partial I_D}{\partial V_{in}}\right) \times R_D \quad G_m = -\frac{\partial I_D}{\partial V_{in}} \Rightarrow A_V = -G_m \times R_D$$

$$G_m = -g_m \frac{1}{1 + g_m R_s} = -\frac{1}{\frac{1}{g_m} + R_S}$$

$$R_s >> 1/g_m \Rightarrow \Delta I_D = \frac{\Delta V_{in}}{R_S} \Rightarrow G_m = \frac{1}{R_s}$$

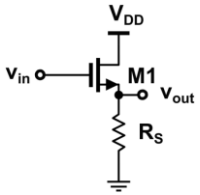



$$A_V = -G_m \times R_D = -\frac{g_m r_o}{R_s + \left[1 + (g_m + g_{mb}) R_s\right] r_o} R_D$$

When $r_O >> R_S$, $A_V \approx -\frac{g_m}{1 + g_m R_s} R_D$

$$R_{out} \approx \left[1 + (g_m + g_{mb}) R_S\right] r_O \approx g_m R_S r_O$$

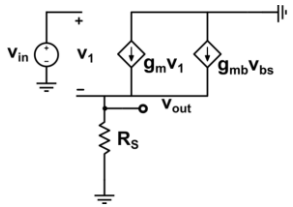
## Common-Drain (Source Follower)



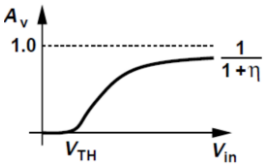

### Large-Signal Analysis

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = \frac{g_m R_S}{1+(g_m+g_{mb})R_S} = \frac{g_m}{\frac{1}{R_S}+g_m+g_{mb}}$$

### Small-Signal Analysis

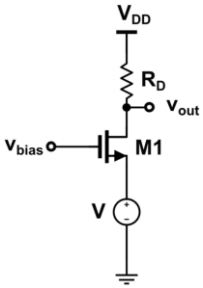



$$A_v = \frac{g_m R_S}{1+(g_m+g_{mb})R_S} = \frac{1}{\frac{1}{g_m R_s}+\left(1+\frac{g_{mb}}{g_m}\right)} \approx \frac{1}{1+\eta} \le 1$$




$$R_{in} = \frac{V_{in}}{I_{in}} \to \infty \text{ (in low frequency)} \quad R_{out} = \frac{1}{g_m+g_{mb}} = \frac{1}{g_m} \Big\| \frac{1}{g_{mb}}$$

## Common-Gate



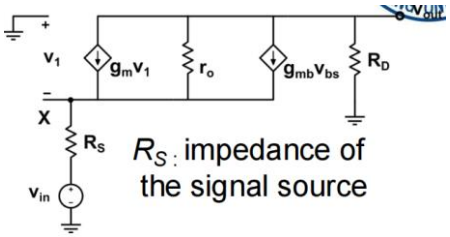

### Large-Signal Analysis

$$\frac{\partial V_{out}}{\partial V_{in}} = \mu_n C_{OX}\frac{W}{L}(V_b - V_{in} - V_{TH})R_D(1+\eta) = g_m(1+\eta)R_D$$

! positive

$$Q\ \frac{\partial V_{TH}}{\partial V_{in}} = \frac{\partial V_{TH}}{\partial V_{SB}} = \eta$$

### Small-Signal Analysis




$$\frac{V_{out}}{V_{in}} = \frac{(g_m+g_{mb})r_O+1}{r_O+(g_m+g_{mb})r_O R_S + R_S + R_D}R_D$$

Input Impedance:

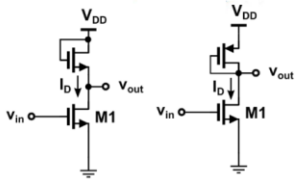
$$\frac{V_X}{I_X} = \frac{R_D+r_O}{1+(g_m+g_{mb})r_O} = \frac{1}{g_O+g_m+g_{mb}}\left(1+\frac{R_D}{r_O}\right)$$

$$\approx \frac{R_D}{(g_m+g_{mb})r_O} + \frac{1}{(g_m+g_{mb})} \qquad \text{if } (g_m+g_{mb})r_O >> 1$$

$$R_{out} = \{[1+(g_m+g_{mb})R_S]r_O + R_S\}//R_D \approx (g_m R_S r_O)//R_D$$

## Summary of CMOS Inverting Amplifiers

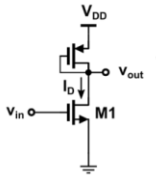
| Inverter | AC Voltage Gain | AC Output Resistance | Bandwidth (CGB=0) |
|---|---|---|---|
| p-channel active load inverter | $\frac{-gm1}{gm2}$ | $\frac{1}{gm2}$ | $\frac{gm2}{CBD1+CGS1+CGS2+CBD2}$ |
| n-channel active load inverter | $\frac{-gm1}{gm2+gmb2}$ | $\frac{1}{gm2+gmb2}$ | $\frac{gm2+gmb2}{CBD1+CGD1+CGS2+CBS2}$ |
| Current source load inverter | $\frac{-gm1}{gds1+gds2}$ | $\frac{1}{gds1+gds2}$ | $\frac{gds1+gds2}{CBD1+CGD1+CDG2+CBD2}$ |
| n-channel depletion load inverter | $\sim\frac{-gm1}{gmb2}$ | $\frac{1}{gmb2+gds1+gds2}$ | $\frac{gmb2+gds1+gds2}{CBD1+CGD1+CGS2+CBD2}$ |
| Push-Pull inverter | $\frac{-(gm1+gm2)}{gds1+gds2}$ | $\frac{1}{gds1+gds2}$ | $\frac{gds1+gds2}{CBD1+CGD1+CGS2+CBD2}$ |



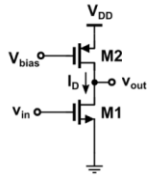

Active NMOS Load Inverter



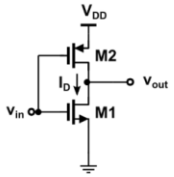

Active PMOS Load Inverter




Depletion NMOS Load Inverter




Current Source Load Inverter




Push-pull Inverter